

Model Paper - Database Management Systems

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Q1 (20 marks)

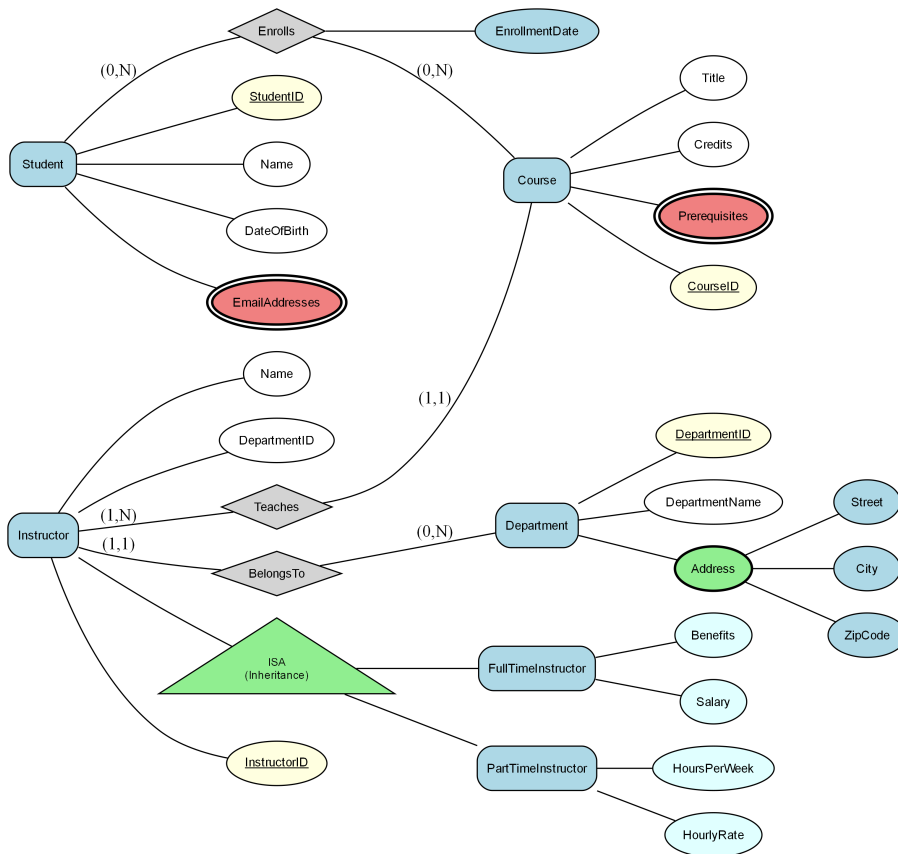
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university system collects comprehensive data on students, courses, instructors, and departments. The Student entity includes attributes such as StudentID, Name, DateOfBirth, and EmailAddresses, while the Course entity has CourseID, Title, Credits, and a multivalued attribute Prerequisites. The Instructor entity has attributes such as InstructorID, Name, Department. features InstructorID, Name, and DepartmentID. The Department entity has attributes such as DepartmentID, DepartmentName, Location. contains DepartmentID, DepartmentName, and a composite attribute Address comprising Street, City, and ZipCode. A relationship 'Enrolls' connects Students to Courses with the attribute EnrollmentDate. Each Course is taught by one Instructor, and each Instructor belongs to one Department. The university employs an ISA hierarchy where the entity The the entity has attributes such as theID, Name, Description. Instructor has subtypes: FullTimeInstructor and PartTimeInstructor, with specific attributes for each subtype. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional). A Instructor can be related to many Courses through the 'Teaches' relationship. Many Instructors can be related to one Department through the 'BelongsTo' relationship (Department participation is optional)

Figure: EER



- Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{ \{AB \rightarrow C, AC \rightarrow D, B \rightarrow E, D \rightarrow A, C \rightarrow B\} \}$.

- Find all the keys in relation R using the attribute closure method. (4 marks)
- Is R in 3NF? Give reasons for your conclusion. (3 marks)
- Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university is redesigning its database system to manage its course offerings, student registrations, and faculty assignments. The database will include tables such as Courses (course_id, course_name, credits), Students (student_id, name, major), and Faculty (faculty_id, name, department). Additionally, relations will be established between the tables to ensure data integrity and optimized queries. The database administrator is tasked with implementing necessary integrity constraints and developing a user-friendly access interface.

- In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement

justifying your answer. (2 marks)

b) Code Segment: `java String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs =
pstmt.executeQuery();` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2 marks)

d) Briefly explain how authentication and authorization are achieved in SQL Server. (3 marks)

e) A university database system is crucial to managing academic information. Identify and discuss three integrity constraints that should be implemented within this database context. Emily is a junior database administrator managing specific databases within the system.. Sarah is the senior database administrator tasked with overseeing the entire database system's creation and maintenance.. Nathan is a database developer responsible for designing and implementing database schemas.. Michael is a data entry operator responsible for inputting data into the system..

- i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Enrollments database using a user-defined server role. (5 marks)
- iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)
- v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a Hospital: Patient (patientId: int, name: varchar(100), address: varchar(150), dob: date) Appointment (appointmentId: int, patientId: int, doctorId: int, appointmentDate: date, status: varchar(20)) Doctor (doctorId: int, name: varchar(100), specialty: varchar(100), phone: varchar(15)) The 'Patient' table stores information about patients including unique patient ID, name, address, and date of birth. The 'Appointment' table contains records of patient appointments with doctors, associating each appointment with a patient and a doctor, along with the appointment date and status. The 'Doctor' table retains information about doctors, including their unique ID, name, specialty, and contact number. The 'Doctor' table manages details about doctors, including unique doctor ID, name, specialization, and contact information. Fee (feeId: int, patientId: int, amount: real, paymentStatus: varchar(20)) The 'Fee' table stores fee/payment details including payable amount and payment status.

a) Write SQL Queries to perform the following: i. Find the name and address of patients who have an appointment scheduled with the doctor named 'Dr. Smith'.

- i. Find the name and address of patients who have an appointment scheduled with the doctor named 'Dr. Smith'. (4 marks)
- ii. Find the patient with the highest aggregate value using GROUP BY on related appointment records. (6

marks)

iii. Find the appointmentDate, status, and name from Appointment, Patient, and Appointment where Appointment.appointmentDate is not null. (7 marks)

b) b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)